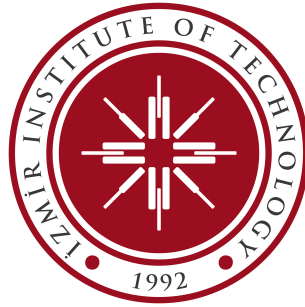




İzmir Institute of Technology
Faculty of Engineering
Department of Electrical and Electronics Engineering

EE 315 Electronics Laboratory Prelab Sheet

Experiment 1

Submitted By:

Baran Güvendiren - 310206029

Ozan İnan - 330206062

Mohammadhossein Nafaryeh - 320206059

October 9, 2026

Problem 1

1.a) When measuring a signal, under what circumstances should each type of voltage value, V_p , V_{pp} , V_{rms} , be used?

- **Peak Voltage (V_p):** Used for determining the maximum voltage levels relative to ground. Example use cases include determining diode's reverse voltage rating or finding the maximum output of a rectifier or clipper. For asymmetric waveforms relative to ground, the positive and negative peaks must be measured separately.
- **Peak-to-Peak Voltage (V_{pp}):** Used when the total swing of the signal is required. It does not depend on the DC offset or the ground reference. For instance, function generators are typically set in terms of V_{pp} . Additionally, it is the standard metric for measuring ripple or noise amplitude.
- **RMS Voltage (V_{rms}):** Used for power calculation and thermal analysis as it represents the equivalent DC voltage which delivers the same average power to load. It is also the best metric for comparing waveforms with different shapes.

1.b) How does the purpose of measurement (e.g., power analysis, amplitude comparison, waveform characterization) influence this choice?

1.c) What are the differences between how oscilloscopes, multimeters, and simulation tools measure RMS or peak values?

1.d) Which limitations (sampling rate, bandwidth, waveform type, numerical filtering, etc.) can affect the accuracy or validity of your selected measurement method?

1.e) Based on your discussion in 1.a), 1.b), 1.c), and 1.d), justify which measurement technique is the most appropriate for measuring V_p , V_{pp} and V_{rms} . Explain your reasoning by referring to the characteristics of each quantity, the type of waveform, and the capabilities of your measurement instruments.

Problem 2

2.a) Show that the RMS value V_{rms} of a sinusoidal voltage waveform $v(t) = V_p \sin(\omega t)$ is given by

$$V_{rms} = \frac{V_p}{\sqrt{2}},$$

Where V_p is the peak value of the voltage waveform.

Problem 3

3.a) Show that the RMS value $V_{\Sigma rms}$ of a voltage waveform $v(t) = V_{DC} + v_{AC}(t)$ is given by

$$V_{\Sigma rms} = \sqrt{V_{DC}^2 + V_{ACrms}^2},$$

Where V_{ACrms} is the RMS value of the pure AC component $v_{AC}(t)$